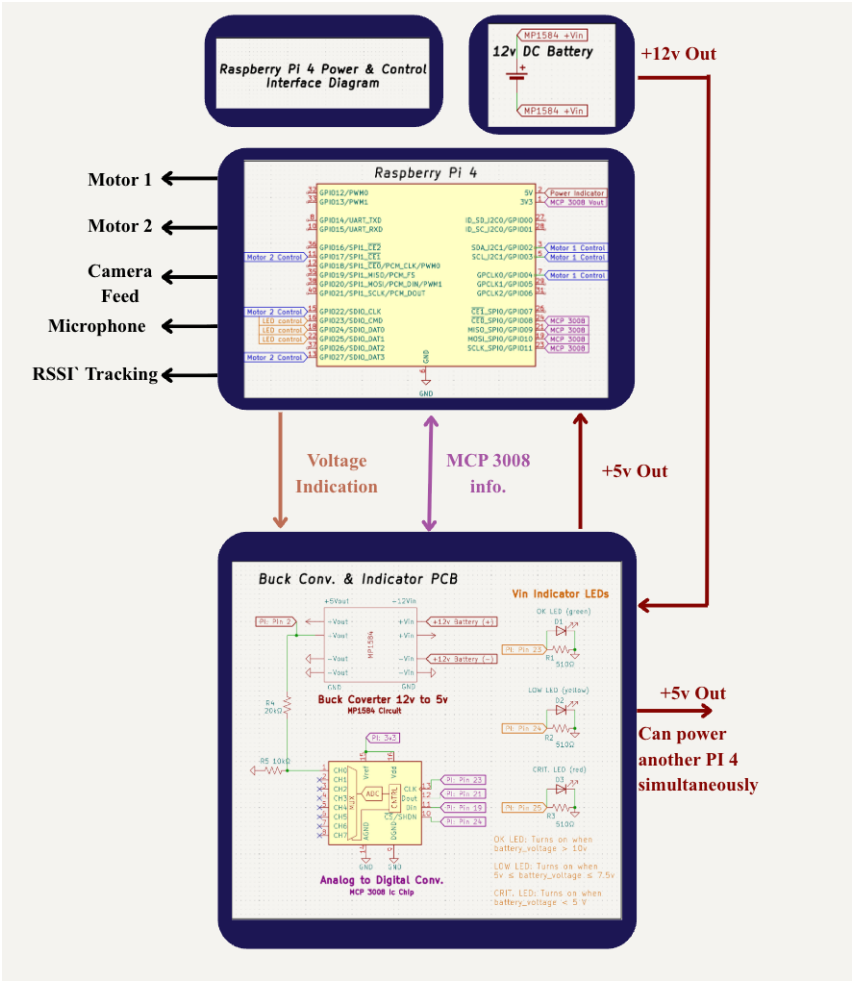
Updated Power and Control System Report

Author: Diego Mendez
Project: BioScan Robot System Functions
Section: Power and Control System
Date: April 2026

System Function Overview

The BioScan Robot is designed to use an onboard Raspberry Pi as the main control unit for system functions such as motor control, camera feed, microphone input, RSSI tracking, and sensor communication. The original system diagram was based on a Raspberry Pi 4 power and control setup, but the design has recently been updated to use a Raspberry Pi 5.

This change is important because the Raspberry Pi 5 requires a stronger and more stable power supply than the Raspberry Pi 4. The current system must provide **5V at 5A** to the Raspberry Pi 5 in order to support reliable operation, especially when multiple sensors and control devices are conected.



Original System Design

The original design used a Raspberry Pi 4 with a buck converter and indicator PCB. The system stepped down battery voltage to provide the required 5V output for the Raspberry Pi. The diagram also included voltage indicator LEDs and an MCP3008 analog-to-digital converter for monitoring battery voltage levels.

In this setup, the Raspberry Pi 4 handled several system functions, including motor control signals, camera feed processing, microphone input, RSSI tracking, and communication with the supporting control board.

Updated Raspberry Pi 5 System

The current version of the BioScan Robot now uses a **Raspberry Pi 5** instead of the Raspberry Pi 4 shown in the original diagram. Because of this upgrade, the power design has also changed.

The Raspberry Pi 5 requires a stable **5V, 5A** power input. This is higher than the Raspberry Pi 4 requirement, so the original power setup is no longer the main design being used. The updated design focuses on giving the Raspberry Pi 5 a more reliable power source to prevent resets, voltage drops, or unstable operation during testing.

Updated Power Source

The main power source has changed from the original testing setup to an **EnginStar Portable Power Station 300W, 296Wh Battery Bank**. This power station includes a **110V pure sine wave AC outlet** and is rated as an **80000mAh backup battery power supply**.

This portable power station is now being used as the main power source for the Raspberry Pi 5 system. It provides a more stable and higher-capacity power option compared to the earlier battery and buck converter testing setup. This also makes the robot easier to test in mobile or outdoor environments because the system is no longer limited to the original small battery setup.

Motor Power Supply

The motors are powered separately from the Raspberry Pi 5 system. The current motor power source is a **9V battery**.

Keeping the motor power separate from the Raspberry Pi power is important because motors can create sudden current spikes when starting, stopping, or changing direction. These spikes can cause voltage drops or electrical noise. By using a separate 9V battery for the motors, the Raspberry Pi 5 is less likely to reset or experience unstable behavior during operation.

Current System Functions

The updated BioScan Robot system supports the following functions:

Motor control signals
Camera feed processing
Microphone input
RSSI tracking
Sensor communication
Voltage monitoring
Power management for the Raspberry Pi 5

The Raspberry Pi 5 acts as the main processing and control unit. It receives data from sensors and connected devices, then sends control signals to the rest of the robot system.

Design Update Note

The current diagram still shows the original Raspberry Pi 4 power and control interface. However, the actual system has recently been updated to use a Raspberry Pi 5. Because of this change, the diagram should be revised in the future to show the new 5V, 5A Raspberry Pi 5 requirement, the EnginStar portable power station, and the separate 9V battery used for the motors. This update improves system reliability by giving the Raspberry Pi 5 a stronger power source while keeping the motor power isolated from the main control electronics.

Design Update Note

The current diagram still shows the original Raspberry Pi 4 power and control interface. However, the actual system has recently been updated to use a Raspberry Pi 5. Because of this change, the diagram should be revised in the future to show the new 5V, 5A Raspberry Pi 5 requirement, the EnginStar portable power station, and the separate 9V battery used for the motors. This update improves system reliability by giving the Raspberry Pi 5 a stronger power source while keeping the motor power isolated from the main control electronics.